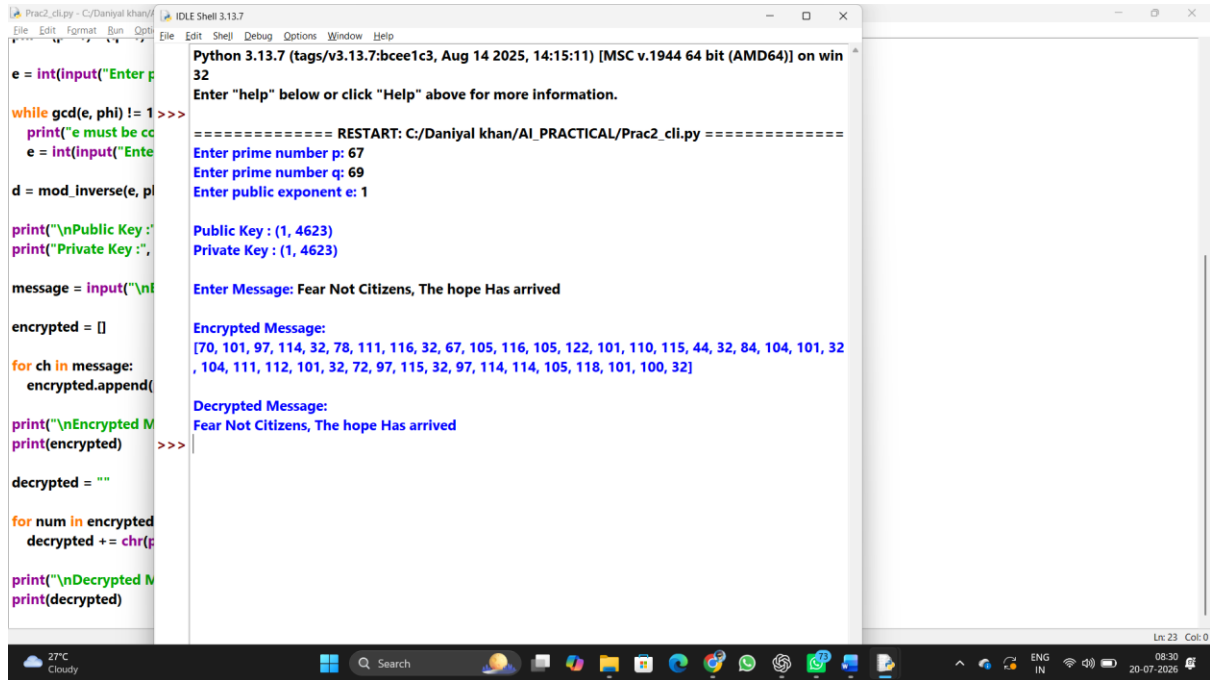


SHETH LUJ COLLEGE

CYBER SECURITY PRACTICAL - 2

Aim:- Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI:Output:-



The screenshot displays a Python IDE window titled 'Prac2_cli.py - C:/Danyal khan/' and a terminal window titled 'Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32'. The code in the IDE implements the RSA algorithm, including functions for calculating the greatest common divisor (gcd), modular inverse, and encryption/decryption. The terminal output shows the execution of the program, where the user enters prime numbers 67 and 69, a public exponent of 1, and a message 'Fear Not Citizens, The hope Has arrived'. The output displays the calculated public and private keys as (1, 4623) and the encrypted message as a list of integers. The decrypted message is shown to be identical to the original input.

```
e = int(input("Enter p: "))
phi = (e-1)*(p-1)
while gcd(e, phi) != 1:
    print("e must be coprime to phi")
    e = int(input("Enter p: "))
d = mod_inverse(e, phi)

print("\nPublic Key : (1, 4623)")
print("Private Key : (1, 4623)")

message = input("\nEnter Message: ")

encrypted = []
for ch in message:
    encrypted.append(ord(ch))

print("\nEncrypted Message:")
print(encrypted)

decrypted = ""
for num in encrypted:
    decrypted += chr(num)

print("\nDecrypted Message:")
print(decrypted)
```

Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:/Danyal khan/Prac2_cli.py =====
Enter prime number p: 67
Enter prime number q: 69
Enter public exponent e: 1
Public Key : (1, 4623)
Private Key : (1, 4623)
Enter Message: Fear Not Citizens, The hope Has arrived
Encrypted Message:
[70, 101, 97, 114, 32, 78, 111, 116, 32, 67, 105, 116, 105, 122, 101, 110, 115, 44, 32, 84, 104, 101, 32, 104, 111, 112, 101, 32, 72, 97, 115, 32, 97, 114, 114, 105, 118, 101, 100, 32]
Decrypted Message:
Fear Not Citizens, The hope Has arrived

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GUI : Output:-

